

CS 217: AI and ML

Weekly Examination

Solutions

Q1

We analyze each statement:

- (A) We move in the direction along negative of the gradient. For any vector v , dot product with a unit vector d is minimum when $d = -\frac{v}{\|v\|_2}$ which is also the desired direction to move along in a gradient descent step.
Hence TRUE.

- (B) For convex functions:

$$F(w) \geq F(w_t) + \nabla F(w_t)^\top (w - w_t)$$

Thus first-order Taylor expansion is a global **lower bound**, not upper bound.

FALSE.

- (C) Even for smooth convex functions, large step sizes can cause divergence. Example: quadratic function with η too large.

TRUE.

- (D) First-order Taylor expansion neglects terms of order

$$O(\|w - w_t\|^2)$$

TRUE.

- (E) Newton's update:

$$w_{t+1} = w_t - \eta_t H_F(w_t)^{-1} \nabla F(w_t)$$

Incorporates curvature information.

But in the exam we forgot to add learning rate. So this option is bonus.

Correct options: (A), (C), (D) / (A), (C), (D), (E)

Q2

If we get anything other than (1,0) for the coefficients, that would mean that the weights we got for the former OLS linear regression were not the optimal ones.

Answer: $39 \times 1 = 39$

Q3

1. TRUE (convex + zero gradient $\Rightarrow$ global minimum)
2. FALSE (Non-convex function might get stuck in local minima)
3. TRUE (Convex function)
4. FALSE (zero gradient may be saddle point)

Correct options: 1, 3

Q4

1. TRUE (unbiased estimator)
2. FALSE (may require more iterations)
3. TRUE (polynomial decay gives convergence)
4. FALSE (uses minibatches)

Correct options: 1, 3

Q5

Check the condition given on 10th slide of Optimization for ML(2) lecture slides

Correct options: 2,3,4

Q6

Data:

$(3, 1), (6, 2), (9, 3)$

OLS estimate:

$$\beta'_1 = \frac{\sum X_i Y_i}{\sum X_i^2} = \frac{3 + 12 + 27}{9 + 36 + 81} = \frac{42}{126} = \frac{1}{3}$$

Variance:

$$\text{Var}(\beta'_1) = \frac{\sigma^2}{\sum X_i^2} = \frac{9}{126} = 0.0714$$

Ridge estimate:

$$\beta'_2 = \frac{42}{126 + 14} = \frac{42}{140} = 0.3$$

Variance:

$$\text{Var}(\beta'_2) = \frac{9 \cdot 126}{(140)^2} = 0.058$$

Bias:

$$E[\beta'_2] = \frac{126}{140}\beta = 0.9\beta$$

Thus:

$$\text{Bias} = -0.1\beta$$

Correct options:

2, 3

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Q7

Incorrect statements:

1. FALSE (Hessian might not exist but function can still be convex)
2. Correct (unbiased)
3. FALSE (Newton is second-order)
4. FALSE (non-convex may have saddle)
5. Correct

Incorrect: 1, 3, 4

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Q8

1. FALSE (first-order uses gradient)
2. TRUE (curvature information)
3. FALSE (may move to saddle/maxima)
4. FALSE (Hessian is matrix)

Correct option: 2

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